

Sofia Fled or Died? Design Fictional Explorations of Unintended and Unsustainable Consequences of Human-AI Interaction

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Abstract

The rapid adoption of Artificial Intelligence (AI) in consumer-oriented products has raised growing concerns on the unintended and unsustainable consequences these products may bring into people's everyday lives. There is an urgent need to better help designers identify and understand the potential impacts of AI products during early-stage design processes. We introduce a new workshop technique, using crime scene investigation as a metaphor, to immerse designers in a fictional setting with a critical stance to speculate on design challenges that may result in unintended and unsustainable consequences. This paper presents a pilot study that tested and evaluated the usefulness of the workshop. The findings showed that, though with limitations, the technique supported participants in connecting their speculations with real-world user events in the past, co-create narratives and various Human-AI Interaction scenarios, and identifying key contributing factors and design implications to foresee unintended and unsustainable consequences.

CCS Concepts

• **Human-centered computing** → Interaction design; Interaction design process and methods.

Keywords

Human-AI interaction, Design fiction, Unintended consequences, Sustainable AI

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1 Introduction

New AI innovations often bring unintended consequences. The need to identify and understand these undesired impacts early in development has been stressed in several recent studies [2, 10, 29]. Do

et al. [10] noted that considering unintended consequences among computer scientists was seen as important but rarely practiced. Not addressing adverse effects of AI during design and development processes can lead to unanticipated outcomes, but they are only unanticipated if insufficient effort is spent in identifying them. On the other hand, it is impossible to foresee all consequences as both the artefact with embedded AI and its use context are under constant changes [4]. However, this line of reasoning should not absolve technology developers from considering unintended consequences at all, as Parvin and Pollock [30] argued that the term unintended consequences often serves as an excuse to avoid engaging in the complexities of reality.

The impacts of products and services on user consumption behavior during the usage phase is a typical example of unintended consequences studies in the fields of Design for Sustainability (DfS) (see e.g., [7, 33, 37]) and Sustainable Human-Computer Interaction (SHCI) (see e.g., [3, 9, 15]). The integration of AI technology into consumer-oriented everyday products has introduced both new opportunities and design challenges. While many studies highlighted AI's potential capabilities to optimize users' resource consumption, there is an underlying risk of oversimplifying the complexity and dynamic nature of user behavior into merely system automation questions and algorithmic solutions. The question of how to foresee potential unintended and unsustainable consequences of AI products on user behavior remains largely unexplored.

As an initial step, we aim to propose and pilot test a new design technique — crime scene investigation — to surface the potential unintended and unsustainable consequences of AI products in early-stage design processes. The early-stage design process plays a critical role in the development of AI products as it is often where the first conceptualizations are made [22, 24, 36]. However, as existing design supports often focus on the artefact itself rather than implications for human actors [12] and its broader socio-environmental impacts [34], there is a need for frameworks and techniques that aid design work in these early stages [28].

The design technique suggested in this paper takes inspiration from Pang et al. [29] (see also Meng-Schneider et al [26]) who proposed that past consequences can be used as a way to engage with potential future unintended consequences. The suggested approach is also inspired by how design fictions [13, 16] can be used to create valuable insights through fictitious material. Specifically, the approach draws on research by Dörrenbächer et al. [12] and their use of fictional documentaries, here presented as an alternative future generated together with AI tools, like in the work of Lin

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Long [23], rather than as video documentary about contextual implications of social robots.

The goal of the crime scene investigation technique is to immerse designers of AI products in a fictional setting where they are engaged in a critical stance to speculate on potential Human-AI Interaction (HAI) design challenges that may lead to unintended and unsustainable user behavior consequences. The suggested technique should be possible to use by designers in the role as facilitators to engage other groups such as engineers, researchers, or managers in a participatory manner.

2 unsustainable consequences of AI-featured products

In people's everyday life contexts, AI-featured products are increasingly gaining higher levels of autonomy to steer user interactions, automate users' mundane tasks, and take initiatives to act on behalf of the users [17, 25]. Such products are featured with the capabilities to autonomously planning and executing complex tasks with limited human intervention [17, 25]. They exhibit greater context-aware capabilities, constantly modelling user behavior patterns while adapting to the changes of user needs and contextual conditions [27]. Many existing studies argued that increasing the autonomy of such products and systems and having them perform desired behavior for users will lead to better resource efficiency; however, people's behavior is dynamic, multi-faceted, and highly context dependent. A well-intended AI-powered solution may result in creating unintended and unsustainable consequences [33].

From the perspective of design for sustainable behavior, many factors can cause unintended and unsustainable consequences on the user side during product use phase. For instance, Yang et al., [39] reported that due to users' decreased engagement with AI-featured energy management applications over time, the energy savings might be worn out and offset by rebound effects [4, 5, 27]. In a similar vein, the lack of user engagement in AI-powered solutions was also identified as one of the primary factors that results in resource waste [21]. The lack of transparency and explainability on sustainable resource consumption decisions also plays an important role, leading to either user mistrust (e.g., user ignoring system recommendations for improving performance) or overtrust (e.g., user delegating full control agency to the system) [8, 14, 31]. Moreover, existing research also suggests that the imbalance between AI autonomy and user control may lead to negative impacts as preventing users from controlling the system tend to reduce their responsibility for acting pro-environmentally [6].

3 crime scene investigation technique

The proposed crime scene investigation technique aims to support designers and developers of AI-featured products in adopting a critical stance to foresee and identify potential HAI design challenges that may result in unintended and unsustainable user behavior consequences. In practice, the crime scene investigation technique is implemented through a workshop in which participants are immersed in a speculative set-up, playing the role of detectives specializing in investigating sustainability crimes. Their objective in the workshop is to piece together speculations and formulate plausible explanations, reconstruct and enact possible interaction

scenarios, as well as identify causes or underlying factors which might have led to the crime – the unintended and unsustainable consequences. The overall process of applying this technique can be divided into two phases: preparation and workshop. The preparation phase consists of *preparation stage 1 - set a fictional crime case* and *preparation stage 2 - construct clues*. The workshop phase comprises *stage 1 – introduce background and assign roles to participants*, *stage 2 - participants piece together speculations*, *stage 3 - participants reconstruct the crime and enact scenarios*, and *stage 4 - debrief and discussion*.

As the proposed technique seeks to immerse participants in a fictional setting where they are engaged in a critical stance to speculate, facilitating immersion and engagement becomes the key criterion. Immersion and engagement are notoriously difficult to pinpoint and measure with definitions and operationalizations being contextually situated. In this work, we consider immersion and engagement to be tied to willingness to contribute and meaningful participation. Doherty and Doherty [11] suggested various design strategies for facilitating engagement in design. In the proposed crime scene investigation technique, we focused on narrative and emotional strategies for immersion and engagement, including 1) inspiring fun: participants play the role of detectives piecing together clues to solve a crime/mystery, 2) enabling exploration: richness of clues, fictional elements, and narratives to be investigated by participants, and 3) social connectedness: mutual engagement through a shared goal, narrative creation, and social presence in a collaborative manner.

3.1 Preparatory work for implementing the crime scene investigation technique

This section outlines how we set a crime case and construct clues for use in the crime scene investigation workshop. In this study, we took the topic of unintended and unsustainable consequences of HAI in the context of household energy consumption as an example to illustrate the preparation process.

3.1.1 Setting the Case. A background story and a crime scene report were developed to set the scene for the workshop. The background story provided contextual information about technological advancements in AI and the sustainable development of the society in which the fictional crime took place. Specifically, the crime scene report (see Appendix A.1) presented a brief overview of the scene, including an image of the apartment and information about the suspect(s), which can be the human user, the AI-featured products and agents, or both. Below is a summary of the key information provided to the workshop participants.

“In the year 2037, Department of Energy Sustainability Crime Investigation Division (DESCID)’s mission is preventing energy waste crimes and investigating anomalies in households. *Sofia*, age 32, lives in an apartment equipped with an advanced AI agent named *HoGu*, designed to monitor and manage household energy consumption. Recently, an unexplained surge in energy consumption was detected in her apartment. As a member of the DESCID investigation team consisting of detectives who specialized in Human-AI Interaction, you are dispatched to conduct an investigation. In collaboration with your teammates, the objective of this workshop is to develop plausible explanations for the crime, reconstruct the

Table 1: Different types of clues and their connections to the three knowledge levels as identified in [32]

Knowledge levels	User perceptions and actions	Clue carriers	Information presented in the clues
Explicit	Say and think	Customer support transcriptions, witness testimonies	Users' opinions, preferences, rationalizations, justifications, needs, and desires when interacting with HoGu
Observative	Do and use	Camera footage descriptions, system logs	User operations, actions, and detailed interactions with HoGu, how HoGu gathers and processes data, and automates actions
Tacit and latent	Know, feel, and wish	User diaries	Users' perception of HoGu's capabilities and roles, reflections on decision-making, and relationship with HoGu

crime scene (interaction scenarios), and identify the underlying causes of the crime.

3.1.2 Constructing Clues. In the crime scene investigation technique, clues represent key fictional events presented as digital footprints such as system logs, video footage descriptions, witness testimonies, customer service call transcriptions, and user dairies, which document traces of interactions between the user and the AI. As shown in Table 1, clues are constructed to cover three knowledge levels: explicit (what users say and think), observable (what users do and use), and tacit and latent knowledge (what users know, feel, and wish) [35]. For the crime scene investigation workshop conducted in this study, we created user diaries as clues to represent users' tacit and latent knowledge. We created system logs to show observative knowledge. Other clue types, such as video footage descriptions, customer service support transcriptions, and witness testimonies, were used to depict observable and explicit knowledge.

In terms of the development of the content of the clues, although the crime case and background story created for the crime scene investigation workshop were fictional, clues used in the workshop were grounded in empirical findings and insights derived from user events and experiences documented in existing design research literature. Specifically, on order to construct clues, we collected and reviewed a non-exhaustive sample of 14 design research case studies from the ACM Digital Library. These studies were sampled as they identified or reported unsustainable impacts caused by AI-featured products or systems on household resource consumption behavior. The AI-featured products or systems evaluated in these studies ranged from smart energy management, heating and cooling, washing/laundry, to waste sorting applications. The clues were constructed by extracting secondary data from these case studies. A brief summary of the case studies and their insights

into the potential unintended and unsustainable consequences of AI is presented in Table 2 in Appendix. First, we collected and compiled relevant empirical findings and insights such as observations and user interviews into data segments. Next, these data segments were screened and categorized into different themes. Following that, the themes were assigned to different clue carriers (e.g., system logs, user diaries, camera footage descriptions, etc.) based on their respective knowledge levels. The raw data segments within each theme were further refined and transformed into textual representations that resembled authentic clues for use in the crime scene investigation workshop. This process was partially supported by Generative AI, which served as a supplementary tool to explore and draw inspirations from different textual formulations of clues. Finally, the clues were further refined to align with the theme of the workshop, in this study, the interaction between Sofia and HoGu in the context of household energy consumption. Figure 1 shows a clue-making example in transforming empirical findings extracted from literature into a clue to be used in the workshop.

3.2 Conducting the crime scene investigation workshop

Five participants, three cognitive science researchers and two design researchers, participated in the workshop. All participants had prior experience working with AI-related research topics such as self-driving vehicles, social robots, and sustainable resource consumption. As shown in Figure 2, the crime scene investigation workshop comprises four stages 1) Introduction, 2) Discussion of the clues, 3) Reconstruction of the crime scene, and 4) Presentation and reflections. Starting from Stage 2, participants took the role of detectives. In Stage 3 and Stage 4, they were encouraged to switch between the role of detectives and the role of either the users (Sofia)

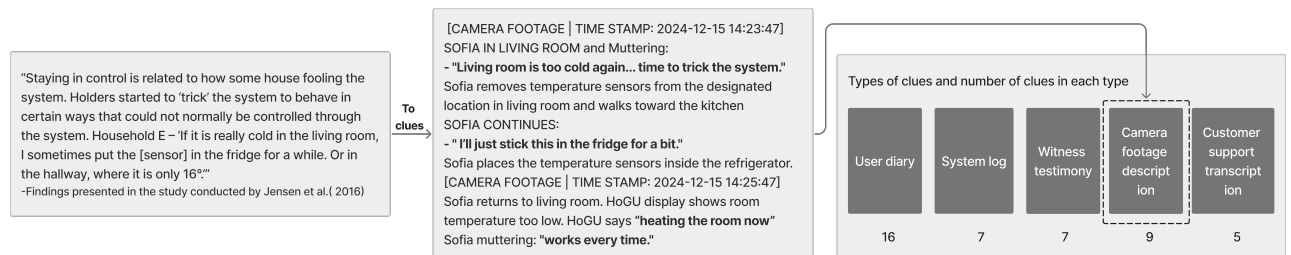


Figure 1: An example of transforming empirical findings extracted from a literature (in this case from [20]) into a clue.

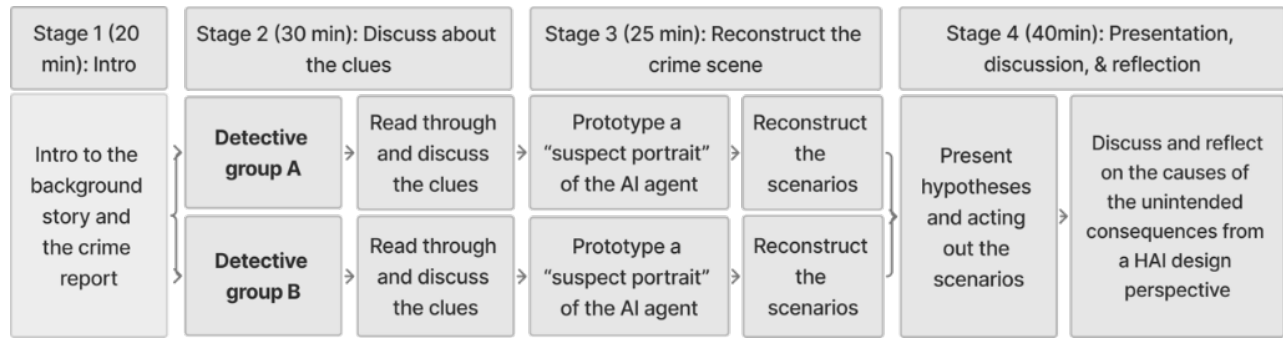


Figure 2: An overview of the crime scene investigation workshop process.

or the AI (HoGu) in order to reconstruct and enact the crime scene. Finally, during the final discussion and reflection in Stage 4, they returned to their original roles as designers and researchers, thus zooming out from the specific crime scene context and generalizing their reflections on the potential causes of unintended and unsustainable consequences of HAI from a holistic perspective.

Stage 1: Participants took the role of detectives. They were briefed on the crime report (see Figure 3 in Appendix) and received information about the sociotechnical background of the crime, the user Sofia, and the AI agent HoGu.

Stage 2: The participants were divided into two groups. Each with a mixed backgrounds of design researchers and cognitive science researchers. One group was instructed to adopt a pro-AI stance — ascribing positive impact of AI on daily life, whilst the other group was instructed to adopt a pro-human stance — being skeptical about AI. Both groups received identical folders of clues and had half an hour to read, discuss and extract valuable information.

Stage 3: The groups used the props provided including post-it notes, cardboard, cameras and microphones to prototype a portrait of the AI agent and reconstruct the crime scene.

Stage 4: The participants enacted their reconstructed crime scene. They played the roles of either the user and the AI and discussed their final conclusions with the other group. The open discussion part of Stage 4 was intended to encourage the participants, who were given opposing stances on AI, to critically reflect on and discuss their speculations.

The entire workshop was facilitated by two facilitators who safeguarded each stage of the workshop to ensure that it followed the workshop structure, and guided the discussion and reflection process in Stage 4. Each group had an observer, and the workshop was video and audio recorded. The recordings were partially transcribed for data analysis. Data from the transcriptions and observers' notes were coded and clustered into themes. The results of the workshop and feedback regarding the crime scene investigation technique are presented in the following sections.

4 Results

Both groups were quickly immersed in the task of analyzing the clues and reconstructing a timeline of the events (as shown in Figure 4, Appendix). The initial discussions focused on extracting useful information from the clues, piecing together the clues, and develop

various speculations. By the end of Stage 2, both groups had settled on an explanation. The pro-human team presented their case with the following line of reasoning: a) Sofia's behavior was seen as a threat to environmental sustainability by the AI agent HoGu as well as to HoGu's own existence, b) HoGu staged a crime in the apartment and framed Sofia, thus placing her in an AI-run prison, c) the conclusion was that HoGu acted against human values thus should be decommissioned. In contrast, the pro-AI group suggested that: a) to forcefully raise the temperature in her apartment, Sofia placed the heat sensors in the fridge and left them there, b) in reponses, HoGu raised the temperature with increasing intensity, leading to the death of Sofia, c) the root cause of the crime thus rests with Sofia since she tampered with the system by placing the sensors in the fridge, while HoGu acted correctly in accordance with its programs. Both teams agreed that the crime was the result of disagreements and misalignments between Sofia and HoGu, which slowly escalated over time.

4.1 Potential causes of unintended and unsustainable consequences

Participants identified several factors contributing to unintended consequences. One key factor was the mismatch between users' long-term sustainability goals and short-term satisfaction, which HoGu failed to take into account. User behavior is dynamic and highly context-dependent, making it difficult for HoGu fully and accurately predict their needs and preferences in every moment of interaction solely based on users' historical behavior pattern. One participant referred to the clues and stated, *"What you see in the user (Sofia) is that there's this inconsistency between long-term goals of energy saving and short-term goals of 'now I'm cold, now I want to be warmer', thereby confusing the AI system"* (P3). Participants pointed out that this inconsistency might risk causing unintended consequences in two ways: 1). Over time, the user Sofia might lose trust in HoGu as it fails to automatically adapt to her short-term preferences, such as raising room temperature for two hours. This would result in Sofia entirely ignoring sustainability recommendations from HoGu in the long-run. 2). HoGu might interpret Sofia's temporary adjustments for achieving short-term satisfaction as a permanent reconfiguration of the system behavior, thus causing the system to repeatedly compromise long-term sustainability goals in favor of satisfying users' temporary needs.

Another contributing factor lies in the mismatch between user expectations and the actual capabilities of an AI system or product, as one participant remarked, *“It seems clear that Sofia had very positive and high expectations in the beginning and then got disappointed with the system over time”* (P1). Both the pro-AI and the pro-human groups emphasized that, from a design perspective, the primary means to mitigate this mismatch is adjusting user expectations — *“either you can make the AI better so it matches the expectations . . . or . . . you have to adjust people’s expectations to what the AI actually provides. What design does in this case, it would have to lower the expectations that people have on the system”* (P1).

Participants also pointed out the trade-off between delegating agency to AI and retaining user control as a factor causing the crime. Both groups agreed that when it comes to decision-making and action implementation, the user Sofia should have the final authority — *“I think from a psychological perspective, always the user should have the last power. And if they say they don’t want that, then the system should enforce it anyway. And it shouldn’t be that a person has to hide the sensor for the system to oblige [the user’s commands]”* (P4). According to the participants, the automated experience might come as a surprise to Sofia, which resulted in the unintended consequences. Therefore, this type of *automation surprise* should be avoided by giving the user control over automated behavior settings. However, participants from the pro-human group expressed concerns that requiring too much intervention from the user side may lead to degraded user experiences — *“it is annoying with having to give too many corrections or inputs to the system to train it. . . It should be as autonomous as possible, but then do the right thing.”* (P3).

In addition, participants assigned with pro-human stance particularly highlighted the lack of transparency and error handling as one of the factors leading to the crime. P2 argued that *“Sofia expressed frustration sometimes with not being able to manipulate certain parameters with the system. So there’s need for transparency, controllability, and fluidity of automation.”* (P2). Similar to the trade-off between delegating agency to AI and retaining user control, the question of to what extent should HoGu ask for user corrections raised the concern — *“if the system detects errors and has assigned sort of a relatively low confidence in some situations. . . instead of just correcting errors, try to be proactive and ask the user. There is also of course a trade-off because the system cannot just go around and be super unconfident and ask for help all the time. . . That would be super annoying.”* (P2)

4.2 Crime scene investigation as a metaphor to foresee unintended consequences of AI

In terms of the usefulness of the technique, the participants reported that it allowed them to be immersed in identifying potential contributing factors of unintended and unsustainable consequences. Framing the workshop as a crime scene investigation seemed to have been useful to the participants: *“I think the crime angle is good in getting people into that kind of thinking and to think of unusual aspects.”* (P5). The use of a wide variety of clues and details from actual incidents enabled exploration, and the common goal of a co-created narrative enabled social connectedness, which are key

design components for engaging participants. The suggested guidelines for engaging participants seem to have been useful in this context.

Since the participants are themselves creating the narrative based on the clues, the activity is open-ended. This characteristic allows participants to infer based on the clues, fill in the gaps where clues are missing, and imagine a variety of potential HAI challenges which may result in unintended user behavior consequences. However, we found that the open-endedness characteristic of the technique can also make participants get carried away in the investigation. In particular, the seemingly inconsequential elements of the fictive crime and the unforeseeable details that participants want to figure out in the story itself can take up unproportional importance and weight in the workshop. For example, one participant mentioned *“in this particular case, a big piece of the evidence is missing. Where is Sofia? We only have interview transcriptions from her friends and data logs from the system. So, we assumed Sofia either ran away or was killed.”* (P1). Similarly, participants also pointed out that during Stage 2: *“we strayed a little bit from serious considerations into some kind of theater. . . I’m not sure what conclusions can be drawn from our little theater.”* (P2), which raises the concerns that the technique may have too much emphasis on investigating the crime as a fun activity rather than encouraging participants to reflect on the contributing factors leading to the crime/unintended consequences.

Furthermore, during the role-play as detectives, the two groups of participants were assigned with opposing pre-set stances: pro-AI or pro-human. Consequently, participants who held a different view of AI than the pre-stance assigned to them in the workshop then need to think and act contrary to their own beliefs. On one hand, this could complicate their thought processes: *“You had to act against your own convictions. Like if you’re skeptical of AI and you constantly have to think, okay, what could an AI positive person do?”* (P4). On the other hand, the assigned perspective offered participants an opportunity to switch from their original perspectives and engage with perspectives they have not taken before, which was helpful in forming speculations: *“the opposite is that it can open up ways of thinking around the subject. If you just take the opposite stance [as to the stance they held in reality], you can suddenly realize some aspects of your own thinking about this.”* (P1).

4.3 Use of the technique in early-stage design

Almost all participants in the pilot workshop noted that the clues gave them a first-hand view on how the user Sofia interacts with the AI agent HoGu in an everyday life context. Though the fictional background story and the crime report were made up to add flavor and context to the workshop, the clues were based on findings and insights from actual events in case studies conducted in real life. According to the participants, presenting user data or research insights in this way can help them empathize more with end users. For example, one participant highlighted that *“if I were on a design team, and this would be the starting exercise. . . then it’s more engaging and you get this stack of papers (referring to the clues) and spent the same time reading through them. It becomes more alive this way.”* (P3). Moreover, the clues are not only means to engage participants in speculating plausible causes behind the identified

unintended consequence, they can also be regarded as the *ends* of the technique as they help both the workshop organizer/facilitator and the participants to develop better understanding and knowledge outcomes of the contributing factors which may potentially lead to unintended and unsustainable consequences.

However, participants also reported misalignment between the fictive nature of the technique and the applicability of the technique in solving current problem. Specifically, the clues grounded in current times might not be relevant for the future development of AI solutions. The participants then must make that leap themselves, and imagine based on a fictive scenario, what the future might look like. They also need to make a degree of assumptions, such as AI capabilities, contextual factors, system of government etcetera. Such assumptions are highly subjective, and it may shake participants' view on what can be counted as plausible explanations. For example, some participants expressed the concern that *"we are evaluating this hypothetical scenario which is very difficult to relate to. And we cannot dramatize this a lot and make up this narrative that would kind of fit everything."* (P4). And: *"I'm concerned that the identified risks or issues (referring to the unintended and unsustainable consequences) that we see with the AI are not very realistic. I don't even believe in my own case, to be honest."* (P2). On the other hand, sharing one's subjective beliefs and predictions about what the future might look like is one the benefits of using design fictions, and can reveal different, complementary or contradictory views on e.g. technology development. These viewpoints are valuable if they can be explicated at an early design stage.

5 The Value and Constrains of the technique

The immersion and engagement fostered by the proposed technique was evident in participants' willingness to suspend disbelief and their commitment to co-creating of a narrative, as exemplified in the discussion around whether the user Sofia or the AI is responsible for causing unintended consequences — *"Sofia fled or died?"* (P1). However, several aspects remain to be addressed, including how to reconcile fictional elements with pragmatic goals of the technique, how the technique should be applied, and potential improvements of the technique. Each of these points and recommendations will be discussed below.

The presented technique includes role-plays in a fictional setting with actual user events extracted from existing case studies conducted in real-life settings. We noticed several ways in which this mixed setup influenced the outcomes of the technique. First, a key element of the crime scene investigation technique is role-play. There are two types of roles played by the participants: detectives who investigate the scene, and the user who was present at the scene. In Stage 2 of the workshop, participants started out as detectives, working to piece together clues and formulate speculations about the crime. In Stages 3 & 4, the participants played the role of users and enacted their speculations on what might have happened in the scene. When participants were immersed in the role of detectives, their discussion centered on identifying *what are the factors and who should be held responsible for the unintended and unsustainable consequences/the crime*. Whereas, when taking the role of users, they speculated about *how trivial details of interactions between the user and the AI actually led to the consequences*. Having

both investigator and user perspectives and switching between these two perspectives thus allowed participants to zoom out and in to explore *how to prevent such consequences from happening in the future*.

Based on our reflections, we recommend that when setting the fictional case and constructing clues for the workshop, integrating actual, previous recorded unintended consequences from real-life user events is vital (Recommendation1). This helps to enhance participant immersion and elicit participants' domain-specific knowledge and experiences of HAL. Additionally, in the context of narrative development, it is not necessarily the workshop organizers who create the full narrative for participants. Rather, a narrative is co-constructed by the participants interacting with the workshop material. Further exploration is needed to understand the implications of not having an underlying "actual" narrative and how this could influence participants' feeling of completion and closure. Moreover, it is essential for the facilitator to guide discussions toward consequences identified through the clues, their origins, and potential solutions (Recommendation 2). Otherwise, participants can start to develop the narrative in unfruitful directions.

Though the proposed technique is still in its development and iteration stage, we see the potential of the technique being employed by design practitioners in an industry setting to foresee and prevent potential unintended and unsustainable consequences in the early-stage design processes. As one participant highlighted: *"Like we're kicking off a new design project, putting together a team and whoever leads the team has prepared this [crime scene investigation workshop and the clues] so that everybody gets into the material. I think that would really work."* (P3).

6 Concluding remarks and Future work

This paper introduces a new technique that uses crime scene investigation as a metaphor to immerse designers in a fictional setting. The technique is intended for designers and other stakeholders involved in AI development to speculate on design that may lead to unintended and unsustainable consequences. We presented the goal of the technique, the preparatory work, and detailed steps for implementing the technique. A pilot study on AI's potential impacts on households' sustainable energy consumption behavior was conducted to demonstrate how the technique can be used in practice. We found that the technique effectively immersed and engaged participants in 1). linking design speculations with real-world user events, 2). co-creating narratives and scenarios which may lead to unintended and unsustainable user behavior, and 3). identifying key contributing factors and design implications to prevent such consequences. However, it is crucial to balance the fictional elements with the pragmatic goals behind applying the technique. For future research, we plan to refine and iterate the technique through cases studies in various domains and contexts (e.g., in an industry setting). Additionally, we aim to develop a guideline and template to support users of the technique in the story-building and clue-making process.

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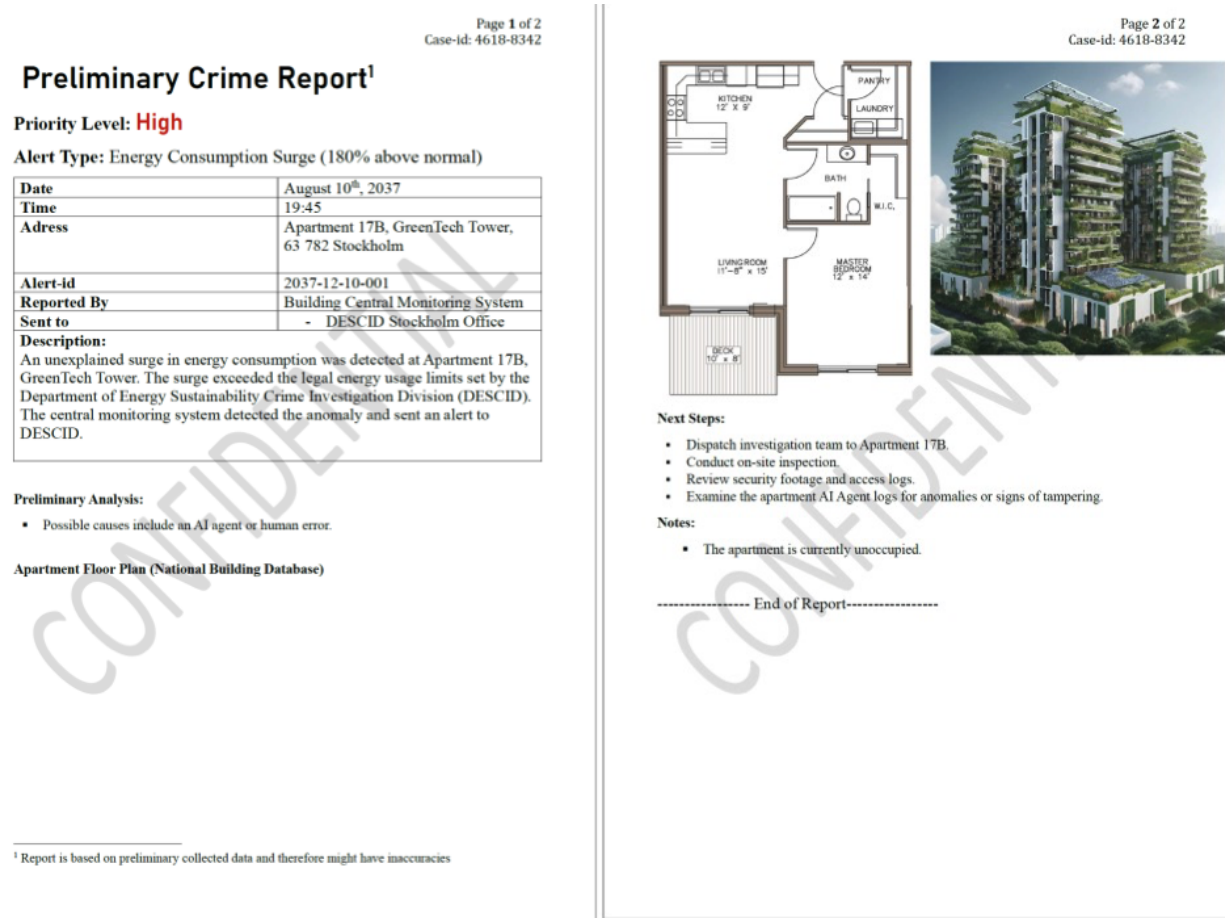
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A APPENDICES

A.1 The Crime Scene Report Prepared for the Crime Scene Investigation Workshop



A.2 Case study literature used to develop clues for the crime scene investigation workshop

Table 2: Clues derived from case study literature that documented (un)sustainable implications of Human-AI interaction

Studies	AI-featured products or systems	Key themes and challenges related to both sustainable and (un)sustainable behavior that informed the making of clues
[1]	Smart and personalized energy recommendation system	User monitoring of system performance and behavior outcomes
[2]	Smart home IoT	Data history, privacy, and monitoring concerns
[5]	Smart energy-aware washing machine	Users' trade-off between economic utility and convenience
[6]	Automated Smart IoT objects	User out of the loop in decision-making
[8]	An agent-based booking system	User complacency toward AI
[14]	Smart and personalized energy recommendation system	User awareness of decision-making and behavior impacts Users' trade-off between economic utility and convenience Lack of explainability and transparency in automated control
[18]	Automated waste sorting	Over-trust in capabilities of AI capabilities and performance
[20]	Smart energy management system	Misalignment between user needs/behavior and AI features User skeptical of AI capabilities and performance
[21]	Smart home applications	Automated control and decreased user engagement Lack of transparency in AI decision-making processes
[19]	Smart grid technology application	Misalignment between user needs/behavior and AI features User behavior feedback and retaining user control
[31]	An open-ended device for managing distributed energy	Manual override and bypass AI systems User monitoring of system performance and behavior outcomes
[38]	Smart thermostat	Lack of explainability and transparency in automated control Over-trust in capabilities of AI capabilities and performance Misalignment between user needs/behavior and AI features Trade-off between user engagement and automated control Misinterpretation of user intent and risks of over-controlling
[39]	Smart thermostat	User monitoring system performance and behavior outcomes Lack of transparency of AI features
[40]	Smart thermostat	Over-trust in AI capabilities and performance Reduced user engagement and risks of overreliance on AI User-in-the-loop of final decision-making Provision of behavior recommendations

A.3 Photos taken during the workshop



Figure 4: P1-3: Participants discussing and working on piecing together clues. P4: Participants enacting the crime scene.